

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks available |     |     |       |       |      |
|----------|-----|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                          | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7/2      | (a) | (i)  |  | <b>A</b> magnesium / Mg (1)<br><b>B</b> carbon dioxide / CO <sub>2</sub> (1)                                                                                                                                                                                                                                                                                                                                                                             |                 | 2   |     | 2     |       |      |
|          |     | (ii) |  | <b>C</b> CuCl <sub>2</sub> (1)<br><b>D</b> H <sub>2</sub> O (1)<br>neutral answers – names of compounds <b>C</b> and <b>D</b>                                                                                                                                                                                                                                                                                                                            |                 | 2   |     | 2     |       |      |
|          | (b) |      |  | <b>2HCl + Na<sub>2</sub>CO<sub>3</sub> → 2NaCl + H<sub>2</sub>O + CO<sub>2</sub></b><br><br>award (1) for correct formula<br>award (1) for balancing<br><br>mark independently – balancing mark can be awarded even if the formula is incorrect                                                                                                                                                                                                          |                 | 2   |     | 2     |       |      |
|          | (c) | (i)  |  | white precipitate / solid (forms)                                                                                                                                                                                                                                                                                                                                                                                                                        | 1               |     |     | 1     |       | 1    |
|          |     | (ii) |  | <div>Ag<sup>+</sup>(aq) + Cl<sup>-</sup>(aq) → AgCl(aq) <input type="checkbox"/></div> <div>Ag<sup>-</sup>(aq) + Cl<sup>+</sup>(aq) → AgCl(aq) <input type="checkbox"/></div> <div>Ag<sup>+</sup>(aq) + Cl<sup>-</sup>(aq) → AgCl(s) <input checked="" type="checkbox"/></div> <div>Ag<sup>+</sup>(s) + Cl<sup>-</sup>(s) → AgCl(s) <input type="checkbox"/></div> <div>Ag<sup>-</sup>(aq) + Cl<sup>+</sup>(aq) → AgCl(s) <input type="checkbox"/></div> |                 |     | 1   | 1     |       |      |
|          |     |      |  | Question 7/2 total                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1               | 6   | 1   | 8     | 0     | 1    |